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Dermal bioavailability of perfluoroalkyl substances using *in vitro* 3D human skin equivalent models

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ABSTRACT

Perfluoroalkyl substances (PFAS) have been identified in various products that come in contact with human skin, ranging from school uniforms to personal care products. Despite this, knowledge on human dermal uptake of PFAS is lacking. Thus, the human dermal absorption of 17 PFAS was assessed, for the first time, using *in vitro* 3D-human skin equivalent models exposed to 500 ng/cm^2 PFAS dissolved in methanol over 24–36 h. The distribution of target PFAS is presented, based on three fractions: absorbed, un-absorbed, and retained within skin tissue (absorbable dose). Perfluoropentanoic acid (PFPeA) and perfluorobutane sulfonate (PFBS) had the highest absorbed fraction, 58.9 % and 48.7 % respectively, with the absorbed fraction decreasing with increasing carbon chain length of the studied perfluorocarboxylic acids (PFCAs) ($r = 0.97$, $p = 0.001$) and perfluorosulfonic acids (PFSAs) ($r = 0.97$, $p = 0.004$). Interestingly, while longer chain PFAS ($C_n \geq 9$) were not directly absorbed, a large fraction of the exposure dose was detected within the skin tissue at the end of the exposure. This was most apparent for perfluoroundecanoic acid (PFUnDA) and perfluorononane sulfonate (PFNS) for which 66.5 % and 68.3 % of the exposure dose was found within the skin tissue, while neither compound was detected in the absorbed fraction. For compounds with a carbon chain length > 11, the fraction found within the skin tissue, decreases with increasing chain length. Physicochemical properties played a role in dermal permeation of PFAS, with a clear inverse correlation between $\log K_{OW}$ and absorbed fraction for both PFCAs ($r = -0.97$; $p \leq 0.001$) and PFSAs ($r = -0.99$; $p \leq 0.001$). Steady-state flux ($\boldsymbol{J_{SS}}$) and permeation coefficients ($\mathbf{P_{app}}$) were determined for target compounds with significant permeation after 36 h exposure (C_5-C_8 PFCAs and C_4-C_7 PFSAs). In general, both the flux and permeation coefficient decreased with increasing chain length.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are a class of synthetic organic chemicals comprising thousands of compounds. Their hydrophobic and lipophobic properties along with their chemical and thermal stability has made them popular for use in a wide range of industries and consumer products (Buck et al., 2011; Glüge et al., 2020). PFAS were first detected in human serum in 2001 (Hansen et al., 2001). Since then, significant human health effects of PFAS have been highlighted i.e., lowered immune response to vaccination (Abraham et al., 2020; Grandjean et al., 2017; Grandjean et al., 2012), impaired liver function (Salihovic et al., 2018) and decreased birth weight (Meng et al., 2018; Sagiv et al., 2018). Additionally, in 2023 PFOA was classified as a group 1 human carcinogen by the International Agency for Research on Cancer's (IARC) (Zahm et al., 2023). Such findings led the European Food Safety Authority (EFSA), to set a recommended tolerable weekly intake of 4.4 ng/kg bw/week for the sum of four PFAS (perfluorooctanoic acid (PFOA), perfluorononananoic acid (PFNA), perfluorohexane sulfonic acid (PFHxS), and perfluorooctane sulfonic acid (PFOS)) in 2020 (EFSA CONTAM Panel, 2020).

Diet and drinking water are considered the main pathways of human exposure to PFAS followed by indoor air and dust (Sunderland et al., 2019). More recently, PFAS have been identified in a wide range of products that come into contact with skin, i.e. cosmetics, hand sanitizers, water-repellent clothing and even school uniforms (Schultes et al., 2018; van der Veen et al., 2020; Whitehead et al., 2021; Xia et al., 2022). Furthermore, significant correlations have been reported between the use of personal care products and PFAS concentrations in both human serum and breast milk (Colles et al., 2020; Kang et al., 2016; Serrano et al., 2021; Thépaut et al., 2021). Additional correlations have been found between serum levels and PFAS detected in hand wipe samples (Kim et al., 2019; Poothong et al., 2019).

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The transfer of compounds across the skin (skin absorption) is a two-fold process. First, the compound enters the *stratum corneum* and the mass of the compound that enters the skin is defined as having undergone dermal penetration. Secondly, the compound permeates through the epidermis and into the systemic circulation (*in vivo*) or the receptor fluid (RF) (*in vitro*). The mass of the compound detected in the receptor fluid is then referred to as that having undergone dermal permeation or dermal absorption. In *in vitro* skin absorption studies, the mass detected within the RF at the end of exposure is referred to as the absorbed, permeated, or bioavailable dose, whereas the mass detected within skin cells after washing at the end of the exposure is referred to as the absorbable dose (OECD, 2004; OECD, 2011; OECD, 2004; Hopf et al., 2020).

A few studies have investigated possible dermal permeation of PFAS (Abraham and Monien, 2022; Chen et al., 2022; Kissel et al., 2023). An *in vivo* study tested the dermal permeation of 15 PFAS in male Sprague-Dawley rats, and found that a fraction of both perfluoroalkyl carboxylates (PFCAs) and perfluoroalkyl sulfonates (PFSAs) were able to penetrate rat skin. The study found that perfluoroheptanoic acid (PFHpA) and perfluorobutane sulfonic acid (PFBS) showed the greatest absorption efficiency with 16 % and 18 % of the applied dose being absorbed respectively, 144 h after exposure. This study also found serum PFAS concentration peaked 8–72 h after dermal exposure, which is significantly later than after oral exposure (1–24 h) (Chen et al., 2022). Another study used a single human subject, exposed to $^{13}C_4$-PFOA mixed into a commercial sunscreen. A peak serum concentration (132 ng/L) was observed 22 days after application supporting the suggestion that dermal exposure can be long-lasting and indeed contribute to total body burdens of PFAS (Abraham and Monien, 2022). Additionally, a paper estimating the dermal absorption of neutral PFAS from the gas phase, identified 11 PFAS, e.g. N-Methyl- and N-Ethyl-perfluorobutyl sulfonamido ethanol (N-MeFBSE and N-EtFBSE) as well as N-Methyl- and N-Ethyl-perfluorohexyl sulfonamido ethanol (N-MeFHxSE and n-EtFHxSE) for which dermal uptake could be a significant pathway compared to inhalation (Kissel et al., 2023).

Although modelling can be a good tool to evaluate the possible dermal uptake of a huge group of chemicals such as PFAS, there are certain limitations in the absence of reliable experimental data. Models are based on physicochemical properties of the compounds, which for surfactants like PFAS can be difficult to determine (Wang et al., 2011). Additionally, when evaluating the possible dermal uptake of compounds from indoor air the hydration of the *stratum corneum* can fluctuate based on the skin's environment which can influence the barrier function of the skin. This could affect the accuracy of the model if not accounted for in its calculations (Weschler and Nazaroff, 2014).

While *in vivo* studies have the benefit of a complete metabolic and physiological system, there are several ethical and technical limitations to such studies, especially those involving human subjects (Ragnarsdóttir et al., 2022). Furthermore, human skin barrier functions can vary based on age, race and skin hydration, making it more desirable to have more than one subject in dermal uptake studies, which in turn raises further ethical questions (Law et al., 2020). While ethical concerns also apply to the use of *in vivo* animal models they remain in use, albeit in decline. However, uncertainties with data derived from animal studies arise due to interspecies variation (e.g., number of hair follicles, intercellular subcutaneous lipids and the skin barrier function) (Abdallah et al., 2015). One study comparing *in vitro* human and rat skin dermal permeation of 14 substances with various physicochemical properties found that human skin was less permeable for all target compounds. No constant factor of difference could be identified (Van Ravenzwaay and Leibold, 2004). In another study which also included an *in vivo* rat model, they found that for 9 out of 12 tested chemicals, *in vitro* human skin was less permeable than the *in vivo* rat skin while the *in vitro* rat skin was more permeable than human skin for all tested compounds (Van Ravenzwaay and Leibold, 2004). A third study found that a higher fraction of tetrabromobisphenol A Bis(2,3-dibromopropyl) ether (TBBPA-BDBPE) permeated through both *in vivo* and *ex vivo* rat skin compared to *ex vivo* human skin (Knudsen et al., 2019). Thus, basing human dermal exposure estimates only on data from rat models can lead to overestimation in risk assessment.

3D-human skin equivalent (3D-HSE) models are commercially available, multilayer dermal tissues that mimic the histological and physiological properties of normal human skin. These models make it possible to conduct dermal absorption studies without the use of human or animal subjects in line with the 3R principles (Ragnarsdóttir et al., 2022). These models were originally developed for the pharmaceutical industry but have been used in previous permeation studies for hazardous industrial chemicals, where they have shown similar barrier function as excised human skin (Abdallah et al., 2015; Schäfer-Korting et al., 2008). A study from 2006 comparing the permeation of three 3D-HSE models, human excised skin and pig skin using 9 different chemicals with varying physicochemical properties found that while the permeation through the 3D-HSE models were higher than those of the *ex vivo* human skin, these models would still be feasible for *in vitro* dermal uptake studies (Schäfer-Korting et al., 2008). Since this study, the 3D-HSE models have been further developed to enhance their barrier function. Using such models listed under the "enhanced barrier function" category, our research group has successfully substituted human *ex vivo* skin with 3D-HSE models to investigate the dermal absorption of various brominated and chlorinated organophosphate flame retardants. While the permeability was slightly increased for the 3D-HSE models compared to excised human skin for some of the tested chemicals, the difference was not significant ($p > 0.05$) (Abdallah et al., 2015; Abdallah et al., 2016).

In the present study, 3D-HSE models (also marketed as having " enhanced barrier function") were applied, for the first time, to investigate the dermal bioavailability of 17 PFAS (10 PFCAs and 7 PFSAs). As this study is the first of its kind, the PFAS included were selected based on the current knowledge on their harm to human health, and the fact that they are the final products of many precursor PFAS (Ragnarsdóttir et al., 2022; Gebbink et al., 2015). Additionally, they have been identified in various products that come in contact with the skin (Schultes et al., 2018; Greenpeace, 2016; Namazkar et al., 2024). We compare the fraction that permeated the skin barrier (dermally bioavailable dose), to the fraction retained within the skin tissue (absorbable dose), and the unabsorbed fraction (that remaining on the skin surface) after 36 h exposure. Finally, the permeation coefficient (K_p), flux (J_{ss}) and lag time (t_{lag}) were determined, providing novel insights into the dermal bioavailability of the studied PFAS.

2. Materials and methods

2.1. Materials

Experiments were conducted according to the principles of good laboratory practice and the OECD guidelines for *in vitro* dermal absorption testing (OECD, 2004; OECD, 2011; OECD, 2004). The study protocol received ethical approval (ERN_19-0779) from the University of Birmingham. Dulbecco's Modified Eagle Medium (DMEM), HPLC grade methanol, ammonium hydroxide, ammonium acetate, Optima LC-MS grade methanol, and Optima LC/MS grade water were obtained from Fisher Scientific (Loughborough, UK). Native and mass labelled PFAS reference standards were purchased from Wellington Laboratories Inc. (Guelph, Canada). Target compounds included $C_{5\text{-}14}$ PFCAs and $C_{4\text{-}10}$ PFSAs (see also table SI-2). PFAS stock solutions (10 ng/μL) were prepared in methanol from reference standards (≥98 %).

2.2. Test matrices

Episkin RHE/L/13 (1.07 cm^2) human skin equivalent kit was obtained from Episkin (Lyon, France). These skin models have shown similar barrier functions and metabolic capacity as *ex vivo* human skin

(Abdallah et al., 2015; Nitsche et al., 2022; Eilstein et al., 2014). The kit includes a maintenance medium (a proprietary DMEM based medium) which allows acceptable tissue differentiation. The supplier instructions of the 3D-HSE models were followed in preparation of the cells before the start of exposure (https://www.episkin.com). Upon receipt, tissues were equilibrated overnight with the supplied maintenance medium at 5 % CO_2 and 37 °C before the start of exposure. Figure SI-1 in the Supporting Information compares the 3D-HSE models and excised human skin.

2.3. Dosing solutions

PFAS stock solutions (10 ng/μL) were prepared in methanol. Following OECD guidelines for infinite dose experiments, 500 ng/cm^2 (50 μL) net loading of each target PFAS was applied to each skin tissue based on the exposed surface area. This ensured the continuous excess of the test compounds in the donor compartment and that the concetrations in the RF did not exceed 10 % of the saturation solubility (OECD, 2004; OECD, 2011; OECD, 2004). Furthermore, this allowed for target PFAS detection at low percentages of the applied dose in the different compartments of the permeation experiment. Finally, the infinite dose approach allows for the determination of steady-state flux(J_{ss}), and permeation coefficients (P_{app}) (OECD, 2004).

2.4. Permeation assay protocol

The infinite dose experiments were performed using the static set-up approach. In an infinite dose application the maximum absorption rate of the test substance is achieved and maintained (OECD, 2004; Hopf et al., 2020). While the concentration is not relevant to real-life human skin exposure conditions, this set-up allows the permeation through the skin to become the rate-limiting step while the concentration gradient is maximized (OECD, 2004). The Episkin tissues were mounted in Franz-type diffusion cells using special inserts constructed for this model with the *stratum corneum* facing up and a DMEM-based culture medium was used as receptor fluid (RF). The DMEM-based culture medium was selected based on the supplier's recommendations. The solubility of the test PFAS was demonstrated according to the OECD guidelines (i.e., 10 times the maximum exposure concentration applied). Furthermore, an addition of 5 % bovine serum albumin (BSA) to the RF is used for highly lipophilic compounds to eliminate solubility in RF as a rate limiting step in *in vitro* dermal uptake tests (as per the OECD guidelines) (OECD, 2011). Initial tests showed no significant change in permeation with the inclusion of 5 % BSA in the receptor fluid and thus the DMEM-based culture medium was used on its own. Figure SI-2 depicts the experimental set-up. All experiments were performed in triplicate. The cells were equilibrated with the RF at 5 % CO_2 and 37 °C for 30 min before target PFAS were applied to the skin surface in the donor compartment. The permeation assay was maintained at 5 % CO_2 and 37 °C throughout the experiment as per supplier instructions. The DMEM-based receptor fluid was sampled (2 mL) at fixed time points (0.5, 1, 2, 4, 8, 10, 12, 14, 16, 19, 22, 24, 28, 32, and 36 h) and immediately replaced with fresh fluid. After 36 h the skin surface was washed thoroughly with cotton buds impregnated with methanol (MeOH, 5 times). The donor compartment was then washed (5 x 1 mL) with MeOH before the tissues were removed from the inserts and stored for analysis. Finally, the receptor compartment was washed with 5 x 1 mL aliquots of MeOH. Each permeation assay generated five different types of samples comprising: receptor fluid (RF) at 15 timepoints and receptor compartment wash, skin surface wash (cotton buds), and donor compartment wash, plus skin tissue. These were then grouped in three major compartments: the *absorbed dose* (cumulative mass in RF + receptor compartment wash), the *skin* ((absorbable dose) mass detected in the skin tissue at the end of the exposure period) and the *unabsorbed dose* (mass in skin surface wash and donor compartment wash at the end of the exposure period). Mass balance was calculated by comparing the sum of recovered analyte masses in all system compartments to the dose applied to the skin at the beginning of exposure. Due to the high number of samples generated, the cost of analysis in the present study and the fact that negligible permeation was detected for C ≥ 9 PFCAs and C ≥ 8 PFSAs after 24 h, no further time-points were collected past 24 h for those compounds. For other target PFAS, initial tests showed no steady-state at 24 h and thus the experiment was extended.

2.5. Sample extraction and chemical analysis

Samples were extracted based on a method previously reported as briefly described hereafter (Harrad et al., 2019). In short, receptor fluid samples (n = 180) were extracted using Chromabond™ PFAS (6 mL/300 mg, Chromabond) SPE cartridges followed by clean-up using EnviCarb™ SPE cartridges. Skin samples (n = 27) were homogenized in 2 mL MeOH followed by 5 min sonication and centrifugation. The samples were concentrated to *ca* 1 mL and PFAS were extracted using the same protocol as receptor fluid samples. Cotton bud samples (n = 27) were extracted with basic MeOH by ultrasonication followed by a clean-up by passing through 0.2 μm syringe filters. Donor and receptor compartment washes (n = 54) were concentrated down using a gentle N_2 flow followed by cleanup using 0.2 μm syringe filters or EnviCarb™ SPE cartridges. A detailed description of sample extraction methods is provided as Supporting Information (SI-2.2.).

Target PFAS were analyzed on a Sciex Exion UPLC coupled to a Sciex 5600 + triple TOF MS operated in negative mode based on a previously reported method (Harrad et al., 2019). Detailed instrumental parameters are provided in Supporting Information (SI-2.3.).

2.6. Quality Assurance/Quality Control

All experiments were performed in triplicate. In addition, a "field" blank comprising of a skin tissue exposed to solvent only but otherwise treated as a sample, was included in each sample batch (n = 3). No target compound was detected in concentrations above the limit of detection (LOD) in the blank samples so no blank correction was needed.

Cell viability was tested using a standard kit purchased from the supplier which included a MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assay. Acceptable MTT results (i.e. Formazan concentration ≥ 1.5 mg/ml) were achieved following 24 h of exposure. Additionally, the integrity of the skin membrane was tested using the standard *trans*-epidermal electrical resistance (TEER) and methylene blue (BLUE) standard methods (Guth et al., 2015). The barrier function of the 3D-HSE models have been tested with the use of positive and negative controls. Triton-X-100 was used as a positive control and showed approximately 100 % permeation, while the negative control (decabromodiphenyl ethane) showed 0 % permeation 24 h after exposure.

Identification and quantification of target analytes were performed according to the relative retention times to the corresponding isotope labelled surrogate standard and the accurate mass of each target PFAS. The robustness of the experimental approach was investigated by calculating the mass balance. Mass balance results revealed generally good recoveries overall (77–113 %, of the target compounds). Further details on QA/QC measures including method accuracy, precision, and limits of quantification for each compound are provided as Supporting Information (Tables SI-1 to SI-3).

2.7. Data analysis

Using Fick's first law of diffusion, a quantitative description of the target compound permeation through the skin barrier was obtained (equation (1) (Niedorf et al., 2008)

$$J_{ss} = \frac{\Delta m}{\Delta t^{*}A} = \frac{D^{*}K^{*}\Delta C}{\Delta x} \quad (1)$$

where J_{ss} = steady-state flux [ng/cm^2*h]; Δm = permeated mass [ng]; Δt = time interval [h]; A = area [cm^2]; D = diffusion coefficient [cm^2/h]; K = partition coefficient; Δc = concentration difference [ng/cm^3]; Δx = thickness of membrane [cm].

When using infinite-dose configurations, where the donor concentrations far exceed the concentrations in the receptor compartment ($C_D \gg C_A$), Δc can be replaced by the known donor concentration, C_D, and the permeated mass per time assumed a constant. Thus, an independent measure of membrane resistance against permeation of the test compound, the apparent permeation coefficient (**P_{app}**) can be calculated (equation (2)

$$P_{app} = \frac{J_{ss}}{C_D} \tag{2}$$

For each compound with a significant permeation after 36 h, cumulative permeation of the compound in the receptor fluid per unit area (ng/cm^2) were plotted against time (hours). Steady state conditions were indicated by a linear regression line ($R^2 \geq 0.9$), where the slope represents the flux (**J_{ss}**). Using a method described by Niedorf et al., 2008 the lower and upper boundary of the linear range (i.e., steady state conditions) was determined (a summary flow chart is provided in the Supporting Information Figure SI-2).

The lag time (**t_{lag}**), with non-detectable flux could then be determined. The lag time represents the time of each compound's partition into and diffusion through the skin and into the receptor fluid following the application of each test compound to the skin models. The **t_{lag}** represents the time intercept (x intercept) of the regression line over the steady-state region of the permeation curve. Thus, **t_{lag}** can be calculated (Equation (3)).

$$t_{lag} = -\frac{b_0}{J_{ss}} \tag{3}$$

where **b_0** is the y-intercept of the linear regression line.

3. Results and discussion

3.1. Dermal bioavailability of PFAS

Results were reported as the average of 3 replicates ± standard deviation (SD). Total recovery calculations revealed that between 77 % and 113 % of the applied dose of PFAS was recovered. Since the target PFAS are not readily metabolised by humans, no metabolism by the skin models was assumed (Cao et al., 2022). The total recovery of PFHxA was lower than that of other compounds. No clear explanation of this could be found and due to lack of time and resources, the permeation experiment could not be repeated for this compound. Table 1 lists the dermally absorbed fractions of target PFAS as well as the PFAS distribution in other fractions in the *in vitro* model compartments at the end of the exposure period.

For target PFCAs and PFSAs, the dermally absorbed fractions decreased with increasing carbon chain length. The absorbed fraction of target PFCAs was highest for PFPeA with over half (58.9 %) of the applied dose being absorbed after 36 h. For PFSAs, PFBS had the highest absorbed fraction (48.7 %), slightly lower compared to PFPeA, yet substantial. Absorbed fractions then decreased for both PFCAs and PFSAs with increasing carbon chain length. In general, PFCAs had higher dermally absorbed fractions after 36 h compared to PFSAs. The compounds with highest absorbed fractions (C ≤ 7 PFCA and C ≤ 6 PFSA) fall under or very close to the Lipinski's criteria of $\log K_{OW} < 5$ and molecular weight < 500 Da (Lipinski et al., 1997). While these criteria were developed for evaluation of drug delivery, our results show good compliance via the observed decreasing dermal absorbed fraction with increasing carbon chain length and $\log K_{OW}$ of the studied PFAS (Fig. 1) (Wang et al., 2011; Chemspider, 2023).

Previous studies have shown that the vehicle chosen for the application of chemicals onto the skin can affect the dermal uptake of those chemicals (Abrams et al., 1993; Chiang et al., 2012). Some organic solvents can delipidate the skin leading to an increased percutaneous absorption of drugs. PFAS amphiphilic properties further complicate the choice of an appropriate vehicle which would not affect the permeability of the skin in any way. A common solvent used in dermal uptake studies, acetone, has for example been shown to increase permeation of amphiphilic compounds (Chiang et al., 2012). Although alcohols could have some skin drying effects which could enhance dermal permeation, they are less occlusive than oil based vehicles which are used to enhance the permeation of lipophilic compounds (Law et al., 2020). Furthermore, PFAS themselves have been used in cosmetics to increase percutaneous permeation (Glüge et al., 2020). Other anionic surfactants such as sodium lauryl sulfate in water also have been shown to increase dermal permeation (Okasaka et al., 2019). Thus, methanol was chosen as a vehicle for the PFAS as they are readily soluble in methanol, and given the low volume used, the solvent evaporated relatively quickly after the single application, leaving the PFAS on the skin surface. Finally, methanol should not affect the skin surface pH and thus its effect on the permeation should be minimal. It cannot however be completely excluded as a factor on the dermal uptake of PFAS which is a limitation of this study.

There is a strong inverse correlation between chain length and absorbed fractions for both PFSAs ($r = -0.97$, $p = 0.004$) and PFCAs ($r = -0.97$, $p = 0.001$). Additionally, functional groups seem to play a role in dermal permeation as PFCAs generally have a higher absorbed fraction compared to the PFSA with the same carbon chain length. These trends were also reported in a recent *in vivo* rat study which included C_4-C_{14} PFCAs (excluding PFTrDA) as well as PFBS, PFHxS and PFOS (Chen et al., 2022). Furthermore, physicochemical properties, e.g., $\log K_{OW}$ seem to influence permeation as there is a clear inverse correlation between $\log K_{OW}$ and permeation for both PFCAs ($r = -0.97$; $p \leq 0.001$) and PFSAs ($r = -0.99$; $p \leq 0.001$). The PFAS mass retained within the skin increased with increasing carbon chain length, peaking for PFUnDA (C = 11) and PFNS (C = 9). The mass retained within the skin then decreased for higher carbon chain compounds. A previous study on the dermal uptake of brominated flame retardants (BFRs) from indoor dust showed similar trends with an inverse correlation between the $\log K_{OW}$ ($p < 0.05$) and the permeated fraction while a positive correlation was

Table 1
Distribution of target PFAS (expressed as % of exposure dose ± standard deviation) in different compartments of the *in vitro* diffusion system after 36 or 24[a] hour exposure to 500 ng/cm^2.

	Absorbed fraction[b]	Skin-epidermis	Unabsorbed fraction[c]	Total recovery
PFPeA	58.9 ± 1.9	17.3 ± 3.5	16.6 ± 2.3	92.8 ± 2.6
PFHxA	36.4 ± 4.1	17.3 ± 3.4	23.5 ± 5.5	77.1 ± 4.4
PFHpA	28.8 ± 3.3	31.8 ± 6.3	32.2 ± 5.9	92.8 ± 5.3
PFOA	13.5 ± 2.8	38.3 ± 5.4	50.9 ± 5.0	102.7 ± 4.6
PFNA	0.9 ± 0.1	67.0 ± 7.3	36.9 ± 9.0	104.8 ± 6.7
PFDA[a]	0.1 ± 0.02	60.7 ± 7.9	33.9 ± 9.4	94.7 ± 7.1
PFUdA[a]	N/A[d]	66.5 ± 11.6	44.5 ± 14.6	111.0 ± 10.7
PFDoA[a]	N/A	31.3 ± 19.5	66.0 ± 18.8	97.2 ± 19.1
PFTrDA[a]	N/A	29.2 ± 1.4	84.2 ± 15.0	113.4 ± 10.7
PFTeDA[a]	N/A	9.0 ± 2.1	77.1 ± 2.4	86.0 ± 2.3
PFBS	48.7 ± 4.2	19.6 ± 4.2	15.9 ± 0.5	84.2 ± 3.4
PFPeS	43.9 ± 2.6	24.4 ± 2.9	25.0 ± 2.4	93.3 ± 2.6
PFHxS	21.9 ± 1.3	33.2 ± 1.5	41.6 ± 3.6	96.8 ± 2.4
PFHpS	6.4 ± 1.3	44.3 ± 1.5	46.5 ± 3.6	97.2 ± 2.4
PFOS[a]	1.3 ± 0.02	60.3 ± 5.0	26.0 ± 5.0	86.6 ± 4.1
PFNS[a]	N/A	68.3 ± 19.6	31.4 ± 5.3	99.7 ± 11.7

[a] PFAS with the last time-point collected at 24 h.
[b] Receptor fluid + receptor compartment wash.
[c] Skin surface wash + donor compartment wash.
[d] Concentration below LOD.

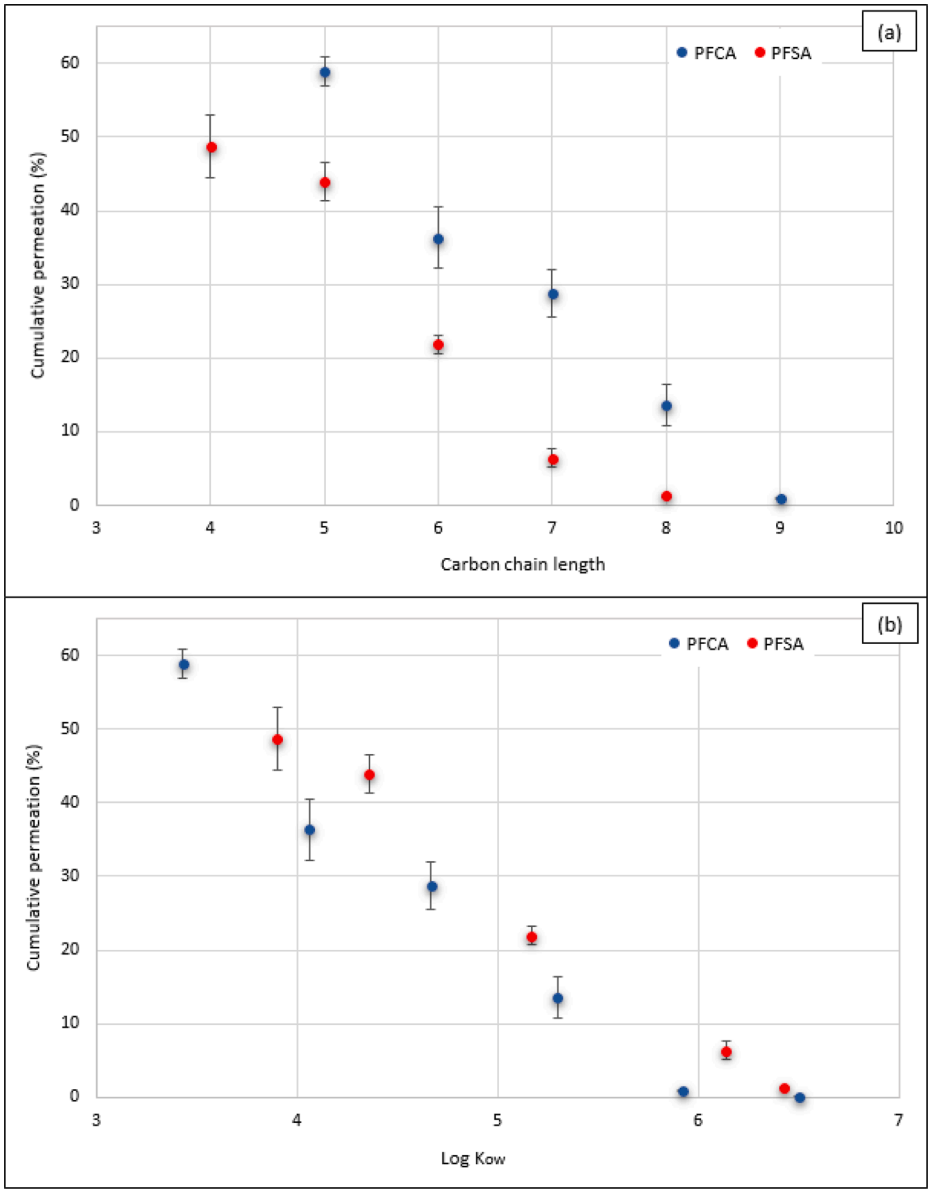

Fig. 1. Association of cumulative permeation of PFAS after 36 h and C chain length (a) and $\log K_{OW}$ (b) Error bars represent the cumulative standard deviation (n = 3). Included are values for compounds with C ≤ 9 and C ≤ 8 for both PFCAs and PFSAs respectively.

seen for $\log K_{OW}$ and the fraction of BFRs retained within the skin ($p < 0.05$). These BFRs are examples of chemicals which do not fit Lipinski's rule of 5 but can still permeate the human skin barrier (Abdallah and Harrad, 2018). The reason for the lower skin retention of the shorter chain PFAS is likely due to their ability to permeate through the skin and enter the receptor fluid. Longer carbon chain PFAS were able to enter the skin cells, and while a steady state was not reached, it may be expected that these long carbon chain PFAS will be less likely to reach the bloodstream but rather be removed by desquamation (Ragnarsdóttir et al., 2022). However, a study on one human subject exposed topically to mass labelled $^{13}C_4$-PFOA mixed into sunscreen revealed absorption was very slow, with plasma levels reaching a maximum (132 ng/L), 22 days after application of a single 110 μg dose (Abraham and Monien, 2022). Furthermore, a study on rats found that the peak times of PFAS in the bloodstream after dermal exposure (8–72 h) were significantly longer than seen after oral exposure (1–24 h), suggesting dermal permeation of PFAS, while slower, could exert a substantial influence on the total body burden (Chen et al., 2022). Keeping this in mind, it could be possible for the fraction of the PFAS still detected in the skin tissue after 36 h to slowly cross the skin barrier and enter the blood stream. Further research is required to understand the potential slow diffusion of long chain PFAS through the human skin barrier. The 3D-HSE model is limited to 36 h as cell viability assays performed by our group as part of the QA/QC protocol revealed > 50 % cell death after 36 h; hence our exposure experiments were limited to 36 h. Possible cell death before 36 h could decrease the number of transporter proteins in the skin cells (Schiffer et al., 2003). It would however not impact the passive diffusion of chemicals through the skin which is considered the main pathway of permeation for most chemicals (OECD, 2004).

Franko et al. investigated the dermal permeation potential of PFOA using *ex vivo* human full thickness skin and epidermis (Namazkar et al., 2024). They saw similar absorbed fractions (24 %) for both skin models after 24 h with an additional 45 % and 23 % of the applied dose found within the skin models. Our findings show a lower permeation of PFOA through the skin (13.5 %) but a similar fraction detected within the 3D-HSE models (38.3 %) and the full thickness skin. There are some differences between the two studies; specifically, the time of exposure 36 h in the current study vs 24 h (Franko et al.), as well as a difference in skin

surface loadings (500 ng/cm^2 in our study compared to 780 x 10^3 ng/cm^2 in that of Franko et al.)), which could explain the differences in the results, especially when expressed as percent of the applied exposure dose. We stress strongly at this juncture that the fraction of the chemical that is permeated is dependent on the applied dose. Consequently, using only the % values for future estimations of dermal uptake from other sources by itself could lead to over-estimations of human exposure to PFAS via dermal uptake (Kissel, 2011; Frasch et al., 2014). Instead, we recommend use of permeation coefficients (P_{app}) as derived and described in the following sections.

3.2. Dermal flux (J_{ss}) and permeation coefficients (P_{app})

A plot of the cumulative absorbed mass of each target compound (ng/cm^2) against time (h) (Fig. 2 a and b) was used to estimate the permeability constant (**P_{app}**, [cm/h]), dermal flux (**J_{ss}**, [ng/cm^2•h]) and lag time (**t_{lag}**, [h]) for each target compound that displayed significant permeation through the 3D-HSE models (Table 2). The steady state range of the curve, presented in Figure SI-4 a and b was identified according to the method of Niedorf et al, with a minimum of 5 data points and the linear range required to establish each curve (Niedorf et al., 2008). The curves show a similar profile for PFCAs and PFSAs with permeation decreasing with increasing carbon chain length. Two compounds, PFBS and PFPeS seem to present a 2-phase kinetic with a first plateau happening between 20 h and 24 h. Further studies are required to better understand the biological significance of this behaviour.

Table 2
Flux (**J_{ss}**, [ng/cm^2•h]), permeation coefficient (**P_{app}**, [cm/h]) and lag time (**t_{lag}**, [h]) values estimated from infinite exposure of 3D-HSE models to 500 ng/cm^2 of the target PFAS with significant permeation after 36 h.

	Flux (Jss) (ng/cm^2 · h)	Permeation coefficient (P_{app}) (cm/h)	Lag time (h)
PFPeA	21.65	$4.33 \cdot 10^{-2}$	0.17
PFHxA	8.99	$1.80 \cdot 10^{-2}$	0.52
PFHpA	4.05	$8.10 \cdot 10^{-3}$	0.14
PFOA	1.91	$3.82 \cdot 10^{-3}$	6.21
PFBS	7.07	$1.41 \cdot 10^{-2}$	0.17
PFPeS	7.01	$1.40 \cdot 10^{-2}$	4.40
PFHxS	3.75	$7.50 \cdot 10^{-3}$	6.85
PFHpS	0.99	$1.98 \cdot 10^{-3}$	5.78

The general trend observed was a decrease in the permeation coefficient (**P_{app}**) with increasing chain length of PFAS. Additionally, the **P_{app}** for PFSAs were lower compared to PFCAs of a similar carbon chain length. Furthermore, an inverse correlation between the estimated **P_{app}**

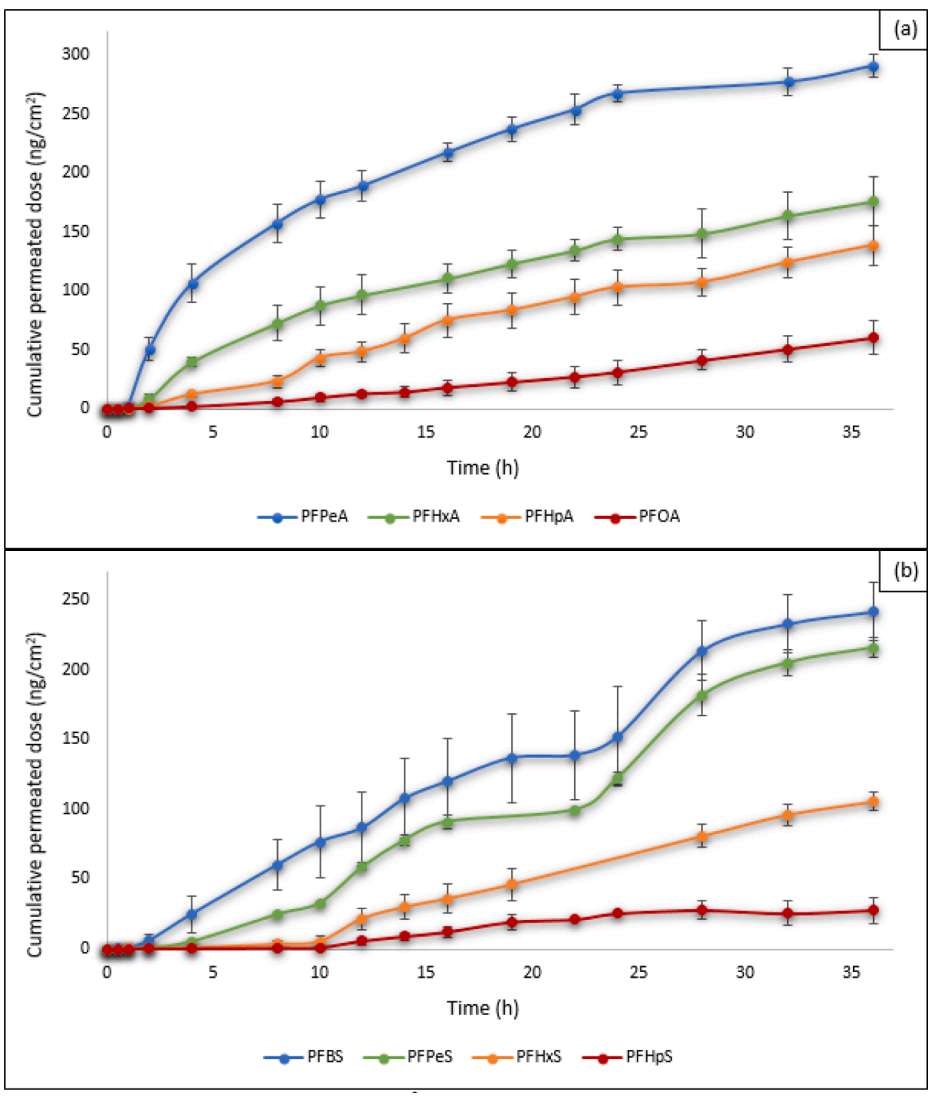

Fig. 2. Cumulative permeation (ng/cm^2) of PFCAs (a) and PFSAs (b) into the receptor fluid following exposure of 500 ng/cm^2 of target PFAS to 3D-HSE models for 36 h. Error bars represent the cumulative standard deviation (n = 3).

of each target PFAS and their $\log K_{ow}$ values ($r = -0.82$, $p = 0.01$) was observed. This trend is in agreement with previous studies of the dermal uptake of halogenated flame retardants where more lipophilic (higher $\log K_{ow}$) compounds (α-HBCDD ($\mathbf{P_{app}} = 2.5 \cdot 10^{-4}$) and TBBP-A ($\mathbf{P_{app}} = 2.9 \cdot 10^{-3}$)) had lower $\mathbf{P_{app}}$ values compared to less lipophilic chemicals i.e. TCEP ($\mathbf{P_{app}} = 2.2 \cdot 10^{-2}$), TCIPP ($\mathbf{P_{app}} = 1.6 \cdot 10^{-2}$) and TDCIPP ($\mathbf{P_{app}} = 0.5 \cdot 10^{-2}$) (Abdallah et al., 2015; Abdallah et al., 2016).

Franko et al. previously estimated the permeation coefficient for PFOA in different ionization states and donor solution pH through *ex vivo* human epidermis (Nitsche et al., 2022). They found (Franko et al., 2012) that for PFOA in the un-ionized form (3000 μg/mL in H_2O, pH = 2.5), $\mathbf{P_{app}} = 5.5 \cdot 10^{-2}$ while the ionized form of the compound (3000 μg/mL in a buffered solution, pH = 5.5) had more than 1000-fold lower $\mathbf{P_{app}} = 4.4 \cdot 10^{-5}$. Our results ($\mathbf{P_{app}} = 3.82 \cdot 10^{-3}$) are 10-fold lower than those reported for the un-ionized form of PFOA by Franko et al. The difference could be associated with a combination of factors including the difference in the dosing vehicle, the skin barrier, exposure time and dosage between the two studies (Frederiksen et al., 2018). It is also worth noting that Franko et al. did not present the curves used for the estimations of $\mathbf{P_{app}}$, so it is unclear whether steady-state conditions were reached in their study.

The ability of other anionic surfactants e.g., sodium lauryl sulfate (SLS) to increase skin permeation has been well studied (Okasaka et al., 2019; Walters et al., 1993; Williams and Barry, 2012). The anionic surfactants themselves can penetrate and interact with the skin altering its barrier properties e.g., by binding to proteins in the *stratum corneum*. Over short exposure periods, anionic surfactants permeate poorly through the skin but at longer application times, permeation increases. While it has been shown that anionic surfactants can permeate the skin, the mechanism of permeation has been debated (Williams and Barry, 2012; Morris et al., 2022; Morris et al., 2019). As SLS is most commonly used in personal care products that are only applied on the skin for short periods of time i.e., soaps and shampoos, most studies focus on short term exposures (Morris et al., 2019; Morris et al., 2019). A longer exposure study however, found SLS in both receptor fluid and within the skin 24 h after exposure. Similarly to PFAS, a steady state was not reached 24 h after SLS exposure (Morris et al., 2019).

3.3. The use of permeation coefficients (P_{app}) for human exposure estimation

Two factors play a role in the process of dermal uptake of a chemical from a source material. In the first step a chemical needs to be released from the source material, i.e., fabric or dust, for it to be available for dermal absorption. This fraction which is released into the body's fluids (in the case of skin this is a mixture of sweat and sebum) is referred to as the *bioaccessible* fraction (Ertl and Butte, 2012; Pawar et al., 2017). This can be a rate limiting step in dermal absorption and has been previously reported for PFAS by our group (Collins et al., 2015; Ragnarsdóttir et al., 2023). The second step in the process would be the transdermal permeation of the *bioaccessible* fraction through the skin and into the systemic circulation. It is not guaranteed that all the *bioaccessible* fraction will reach the systemic circulation. The fraction that does reach the systemic circulation is referred to as the *bioavailable* fraction (Ertl and Butte, 2012; Ragnarsdóttir et al., 2023). In the current study, only PFAS dermal bioavailability is presented while an *in vivo* study accounts for both steps. Furthermore, the dose dependency of the permeated fraction of a chemical is well established, and dermal uptake values at one dose concentration expressed as a percentage of that dose are not necessarily transferable to other conditions such as different dose concentrations (Kissel, 2011). Thus, we present here permeation coefficients (P_{app}) which are not dose dependent and which we recommend for use in human exposure studies.

To demonstrate the use of the permeation coefficients, the dermal permeation of PFOA was estimated in the same scenario as presented in a single human dermal exposure by Abraham and Monien and the estimated results compared to those presented in that study (Abraham and Monien, 2022). These calculations can be found in the Supporting Information (section SI-1, equations SI-1 to SI-5). Using the $\mathbf{P_{app}}$ presented here resulted in a body burden of 21 ng/kg bw. When also accounting for the *bioaccessibility* from indoor dust estimated by our study previously (70 % (Ragnarsdóttir et al., 2023), this resulted in a body burden of 14.7 ng/kg bw.

The same body burdens were then estimated based on the values identified in plasma and presented in Abraham and Monien (Abraham and Monien, 2022). This resulted in a body burden of 2.8 ng/kg bw 36 h after exposure. However the peak plasma levels were seen 22 days after exposure. Thus, the 4 h exposure to PFOA in sunscreen performed by Abraham and Monien resulted in a peak body burden of 13.1 ng/kg bw.

When comparing the estimated results based on the permeation coefficients (21 ng/kg bw or 14.7 ng/kg bw (bioaccessibility included)) and the peak plasma values presented by Abraham and Monien (13.1 ng/kg bw), the difference between the two are not too great especially when the bioaccessibility is also accounted for. Unfortunately, the one sample precludes meaningful statistical testing of this difference. Nevertheless, the plasma concentrations reported by Abraham and Monien only account for the PFOA in that one compartment – the plasma. No other possible fate of the PFOA, e.g., excretion, organ accumulation, or what is remaining within the skin tissue itself could be accounted for. Furthermore, an interindividual variation (skin function variation based on age, race and skin hydration etc. (Law et al., 2020) can not be accounted for based on the study by Abraham and Monien as the study included only one volunteer. The model presented here however, provides an estimate of all PFOA that permeated through the skin, regardless of the tissue/organ where it accumulated in the human body and regardless of the amount excreted after time has passed after the exposure. This could explain the slightly higher values resulting in the $\mathbf{P_{app}}$ estimations compared to the values presented from the *in vivo* study. A further uncertainty is that the bioaccessibility of PFOA from sunscreen was assumed to be identical to that from indoor dust. The present study is in line with the EU's directive for the protection of animals used for scientific purposes and the 3R principle (Replacement, Reduction and Refinement). This subsequently led to massive efforts towards the integration of New Approach Methodologies i.e. *in vitro* studies using 3D-HSE models like this one for risk assessments studies (Pistollato et al., 2021; Barthe et al., 2021). Keeping this in mind in addition to the similar results to the *in vivo* human study, further highlights the the importance of the study presented here.

4. Conclusion

The current study provides information on the dermal bioavailability of 17 PFAS. Substantial dermal absorption (>5% of exposure dose) was seen for $C \leq 8$ PFCAs and $C \leq 7$ PFSAs. The shortest carbon chain compounds examined, PFPeA (58.9 %) and PFBS (48.7 %) had the highest absorbed fractions of PFCAs and PFSAs respectively, with absorbed fractions of PFAS decreasing with increasing carbon chain length. A previous study showed pH can impact the dermal penetration potential of PFAS (Franko et al., 2012). While PFAS are more likely to be present in the ionized form on the skin surface (e.g., in personal care products), it has been shown that they are still capable of permeating the skin (Abraham and Monien, 2022). Similarly, other anionic surfactants are also able to penetrate the skin (Morris et al., 2019). Based on the data presented in this study, dermal exposure could be a significant source of exposure for some PFAS, especially the shorter-chain PFAS. Thus, the dermal route should not be dismissed as a possible route of human exposure to PFAS.

Finally, the permeation coefficients ($\mathbf{P_{app}}$) provided here can be used to estimate human dermal uptake of PFAS in short term exposure scenarios. To the best of our knowledge this is the first time such coefficients are presented for seven out of the eight PFAS for which permeation coefficients were estimated in this study (PFOA excluded).

These permeation coefficients should aid in future human dermal exposure estimation models. Further studies that mimic real-life exposure scenarios would further assist the current understanding of the relative contribution of the dermal exposure pathway to human body burdens of PFAS, compared to other, well-established exposure routes.

CRediT authorship contribution statement

Oddný Ragnarsdóttir: Writing – original draft, Visualization, Investigation, Data curation. **Mohamed Abou-Elwafa Abdallah:** Writing – review & editing, Supervision, Methodology. **Stuart Harrad:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2024.108772.

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